Saeid Zare, Ph.D.

Senior R&D Engineer: thermal-fluid systems, CFD and combustion

Bellefonte, PA | saeidzare.me@gmail.com
linkedin.com/in/saeidzare89 | scholar.google.com (Saeid Zare) | U.S. Permanent Resident

Summary

Mechanical engineer (Ph.D.) working across thermal-fluid systems, heat transfer, combustion and CFD, and across the whole R&D cycle, from simulation and design through testing to customer support. At Faber Burner Company since 2021, now Senior R&D Engineer: developed two burner product families, improved burner NOx performance, expanded the test facilities, and co-invented three U.S. patent filings, one granted. Before that, managed a university combustion laboratory, carried out NASA-funded doctoral research, and taught thermodynamics and heat transfer. Author of 7 journal articles cited more than 170 times.

Areas of Expertise

Thermal-fluids	Heat transfer, thermodynamics, fluid mechanics, combustion, burner design, emissions (NOx) reduction
Simulation	CFD (Ansys Fluent, OpenFOAM), chemical kinetics (Cantera), design of experiments, uncertainty analysis, statistics
Testing	Test-facility development, test procedures, PLIF (including system operation and maintenance), PIV, schlieren and OH chemiluminescence imaging
R&D leadership	Product development, patents, project planning (PERT, MS Project), team supervision, technical communication with customers and marketing
Software	SolidWorks, CATIA, Pro/ENGINEER, Python, MATLAB, Fortran

Experience

Senior Research and Development Engineer May 2025 – present
Burner Development/Design Engineer May 2021 – May 2025
Faber Burner Company, Lock Haven, PA

- Developed two burner product families, taking each from concept through CFD simulation and combustion testing.
- Improved the NOx performance of the company's burners.
- Ran Ansys Fluent CFD simulations and burner combustion tests to evaluate the flow, combustion and emissions performance of new designs.
- Expanded the R&D test facilities and improved the R&D test procedures.
- Co-inventor on three U.S. patent filings on burner and low-emission combustion designs, including U.S. Design Patent D1,102,821 (granted 2025).
- Built burner-sizing software that engineering and marketing use to select burners for customer applications.
- Developed customer support programs and updated the technical communication sheets sent to customers.
- Designed and launched the company's new website.

Graduate Research Assistant and Laboratory Manager Aug 2017 – Aug 2021
Mississippi State University, Mississippi State, MS

- Managed the Plasma and Combustion Research Laboratory and supervised seven graduate and undergraduate students.
- Carried out a NASA-funded study of nanosecond pulsed plasma discharges for stabilizing inverse diffusion flames, the basis of the dissertation.
- Studied plasma-assisted ignition and discharge mode transition in a constant-volume combustion chamber, and the laminar burning speed of biofuel blends (anisole, oxygenated gasoline additives).
- Used schlieren, PLIF, PIV and OH chemiluminescence imaging, paired with chemical-kinetics and CFD modeling (Cantera, Ansys Fluent).
- Wrote research proposals, and published the work in the journal articles and conference papers listed below.
- Instructor of record for Thermodynamics II (three semesters, 130 students in online and hybrid sections) and Intermediate Heat Transfer (graduate and undergraduate); teaching assistant for Internal Combustion Engines.

R&D Researcher and Commercial Manager 2015 – 2017
Avin Exon Pars Ltd., Mashhad, Iran

- Designed product accessories to customer requirements, sourced vendors and prepared sales plans.

Aerodynamicist 2010 – 2011, 2013 – 2016
Sun and Air Research Institute, Mashhad, Iran

- Conceptual design of a two-seat ultralight aircraft, performance and flight-dynamics analysis, wind-tunnel testing, and FAA and JAA certification standards.

Graduate Research Assistant 2012 – 2016
Micro/Nanofluidics Lab, Ferdowsi University of Mashhad, Iran

- Drop-impact and fluid–fluid interface experiments using planar laser-induced fluorescence (PLIF).

Founding Member and Engineer 2011 – 2014
Hakat Gostar, Iran

Education

Ph.D., Mechanical Engineering 2021
Mississippi State University, Mississippi State, MS | GPA 4.0/4.0
Dissertation: *Experimental Study of Laminar Burning Speed and Plasma-Stabilized Flame*

M.Sc., Mechanical Engineering (Energy Conversion) 2016
Ferdowsi University of Mashhad, Iran
Thesis: *Experimental Study of Drop Impact and Penetration into a Water Surface Using PLIF*

B.Sc., Mechanical Engineering (Thermo-Fluids) 2012
Ferdowsi University of Mashhad, Iran
Thesis: *Aerodynamic Study of Bird Flight Applied to the Cinereous Vulture*

Patents

[P1] **S. Zare**, T. A. Probst, J. Colannino, “Diffuser plate,” U.S. Design Patent D1,102,821 S, issued November 25, 2025 (filed July 28, 2023); assigned to Faber Burner Company.

[P2] J. Colannino, **S. Zare**, T. A. Probst, “Combustion system with airflow eductor for pollution reduction,” U.S. Patent Application No. 19/237,972, filed June 13, 2025; assigned to Faber Burner Company.

[P3] J. Colannino, **S. Zare**, T. A. Probst, “Combustion system with mixing and flame arresting for pollution reduction,” U.S. Patent Application No. 18/512,459, published 2024.

Journal Articles

[J1] **S. Zare**, N. Druker, J. Lefkowitz, O. Askari, “Experimental and numerical study of stabilized flame in inverse coflow turbulent jet using nanosecond repetitively pulsed discharges,” *Combustion and Flame* 266 (2024) 113515.

[J2] **S. Zare**, O. Askari, “Mode transition and uncertainty analysis of repetitive nanosecond pulsed discharge,” *Journal of Electrostatics* 118 (2022) 103736.

[J3] J. Shaffer, O. Askari, **S. Zare**, “Electrode design for thermal and nonthermal plasma discharge inside a constant volume combustion chamber,” *Journal of Energy Resources Technology* 144(8) (2022) 082306.

[J4] **S. Zare**, H. W. Lo, S. Roy, O. Askari, “On the low-temperature plasma discharge in methane/air diffusion flames,” *Energy* 197 (2020) 117185.

[J5] **S. Zare**, H. W. Lo, O. Askari, “Flame stability in inverse coaxial injector using repetitive nanosecond pulsed plasma,” *Journal of Energy Resources Technology* 142(8) (2020) 082101.

[J6] **S. Zare**, S. Roy, A. El-Maadi, O. Askari, “An investigation on laminar burning speed and flame structure of anisole-air mixture,” *Fuel* 244 (2019) 120–131.

[J7] S. Roy, **S. Zare**, O. Askari, “Understanding the effect of oxygenated additives on combustion characteristics of gasoline,” *Journal of Energy Resources Technology* 141(2) (2019) 022205.

Conference Papers

[C1] J. Shaffer, **S. Zare**, O. Askari, “Modeling discharge spark ignition using zero dimension thermodynamic model and experimental power measurements at various pressures,” *ASME International Mechanical Engineering Congress and Exposition (IMECE2021)*, 2021.

[C2] J. Shaffer, **S. Zare**, O. Askari, “Structure and measurement of atmospheric and high-pressure ignition plasma,” *ASME IMECE2021*, 2021.

[C3] **S. Zare**, O. Askari, “Instability study of repetitive nanosecond pulsed discharge plasma in a plasma assisted burner,” *ASME IMECE2020*, 2020.

[C4] **S. Zare**, H. W. Lo, S. Roy, O. Askari, “Flame stability in inverse coaxial injector using repetitive nanosecond pulsed plasma,” *ASME IMECE2019*, Salt Lake City, UT, 2019.

[C5] S. Roy, **S. Zare**, O. Askari, “Understanding the effect of oxygenated biofuels addition on combustion characteristics of gasoline,” *ASME IMECE2018*, 2018.

[C6] **S. Zare**, S. Roy, L. A. Onigbanjo, O. Askari, “Experimental and numerical investigation of flame kernel formation and propagation in methane/air mixtures,” *Spring Technical Meeting of the Central States Section of the Combustion Institute*, Minneapolis, MN, 2018.

[C7] **S. Zare**, M. Passandideh-Fard, “Experimental study of water drop penetration on water surface using planar laser induced fluorescence,” *24th Annual International Conference on Mechanical Engineering (ISME2016)*, Yazd, Iran, 2016.

[C8] **S. Zare**, M. Passandideh-Fard, “Experimental study of water drop penetration following drop impact on water surface using planar laser induced fluorescence,” *16th Conference on Fluid Dynamics (FD2015)*, Iran, 2015.

Conference Presentations

[T1] “Experimental study of repetitive nanosecond pulsed discharge on inverse diffusion flame,” *Spring Technical Meeting of the Central States Section of the Combustion Institute*, 2020.

[T2] “Understanding the effect of nanosecond pulsed discharge on ignition and flame stability of methane jet flames,” *11th U.S. National Combustion Meeting*, Pasadena, CA, 2019.

[T3] “On the laminar burning speed and spherical flame structure of anisole-air premixed mixture,” *11th U.S. National Combustion Meeting*, Pasadena, CA, 2019.

[T4] “Effect of capacitive discharge ignition on plasma combustion of propane-air mixture,” *11th U.S. National Combustion Meeting*, Pasadena, CA, 2019.

[T5] “Development of a new chemical mechanism for ethanol-air mixture in a wide range of temperature and pressure,” *11th U.S. National Combustion Meeting*, Pasadena, CA, 2019.

[T6] “Experimental study on laminar burning speed for anisole/air mixture,” *Spring Technical Meeting of the Central States Section of the Combustion Institute*, Minneapolis, MN, 2018; also an arXiv preprint (2018).

Honors and Awards

Preparing the Future Professoriate Program Award, Bagley College of Engineering, Mississippi State University ($21,600; one of two Ph.D. candidates selected)	2019
Award for Technical Merit, 11th U.S. National Combustion Meeting	2019
Award for Artistic Merit, Spring Technical Meeting of the Central States Section of the Combustion Institute	2018

Certifications

Google AI Essentials, Google	2025
Introduction to Statistics, Stanford University	2025
Python Data Structures, University of Michigan	2024
Programming for Everybody (Getting Started with Python), University of Michigan	2024
Applied Computational Fluid Dynamics, Siemens	2023